Question

Element **X** is mainly derived from associated minerals such as molybdenite. When molybdenite is roasted, **X** is converted to the volatile oxide **A**. Dissolving **A** in ammonia yields salt **B**, which can be reduced with hydrogen to obtain **X**. Reducing **A** with CO produces oxide **C**, whose crystal is cubic and has very low resistivity. Spongy **X** reacts with chlorine at $400 \sim 500^{\circ}\mathrm{C}$ to obtain **D**, and **D** disproportionates upon contact with water, forming **E** and oxide **F** in a 1:2 ratio. The oxidation state of **X** in **E** is the same as in **A**. The structure of **D** is actually a dimer. In a nitrogen atmosphere, **D** undergoes thermal decomposition to produce **G** (containing **X** 63.65%) and gas **H**.

Deduce the substances represented by the above letters and determine the correctness of the following statements.

1. **X** belongs to Group 8

2. The valence of the metal in **A** is greater than or equal to 6

3. The formula weight of **B** is greater than 250

4. The low resistivity of **C** is due to the presence of metallic bonds in the crystal

5. **D** is stable in air

6. The formula weight of **G** is greater than 300

7. **H** can be used for water purification

Let a be the sum of the squares of the serial numbers of all correct statements, and b be the square of the sum of the serial numbers of all incorrect statements. Find the value of a/b, taking three significant figures.

A. All other options are incorrect

B. 0.242

C. 0.542

D. 0.280

E. 2.10

F. 0.147

Answer

Correct Answer: **B**

Detailed Explanation

The entry point of this question is **G**. **D** is obtained from the reaction of **X** element and chlorine gas, and is a chloride of **X**. **D** decomposes upon heating to yield **G** and gas **H**. It is evident that **G** is also a chloride of **X**, while **H** is a chloride of **X** or chlorine gas. Let **G** be XCl_n. Given that **G** contains **X** at 63.65%, calculate the atomic weight of **X**. Only when n=3 does the atomic weight of **X** equal 186.2, corresponding to Re. All other calculation results do not correspond to any element. **G** is $ReCl_3$.

CHECKPOINT 2 PTS

X is Re

CHECKPOINT 1 PTS

G is $ReCl_3$

Re is an element of the seventh subgroup, so statement 1 is incorrect.

The formula weight of **G** is 293, which is less than 300, so statement 6 is incorrect.

Calcination of a mineral containing **X** yields a volatile **A**. From common elemental knowledge, **A** is Re_2O_7. When **A** is dissolved in ammonia water, **B** obtained is NH_4ReO_4. The common preparation method for rhenium is the reduction of ammonium perrhenate with hydrogen gas, which is consistent with the description in the question.

The valence state of the metal element in **A** is 7, which is greater than 6, so statement 2 is correct.

The formula weight of **B** is 268, which is greater than 250, so statement 3 is correct.

CHECKPOINT 1 PTS

A is Re_2O_7

CHECKPOINT 1 PTS

B is NH_4ReO_4

C is obtained by reducing **A** with CO and is a low-resistivity oxide with a cubic crystal system. Among the oxides of rhenium, only the classic ReO_3 possesses such properties. Its crystal structure can be viewed as rhenium atoms at the vertices of the unit cell, with oxygen atoms occupying all the edge centers. Its low resistivity stems from the spatially delocalized electron network formed by the conjugation of rhenium atoms through oxygen atoms. Therefore, statement 4 is incorrect.

CHECKPOINT 1 PTS

C is ReO_3

CHECKPOINT 1 PTS

The low resistivity of ReO_3 is due to the formation of a delocalized electron network through the conjugation of metal atoms through oxygen atoms in the crystal

Chloride **D** disproportionates upon contact with water, forming **E** and oxide **F** in a 1:2 ratio. The oxidation state of **X** in **E** is the same as in **A**, so **E** is $HReO_4$. There is water vapor in the air, so **D** cannot exist stably in the air, and statement 5 is incorrect.

Based on the metal valence state and the disproportionation ratio, there are only two possibilities: either **D** is $ReCl_5$ and **F** is ReO_2, or **D** is $ReCl_3$ and **F** is Re_2O. However, **G** is already $ReCl_3$. Therefore, **D** is $ReCl_5$ and **F** is ReO_2.

CHECKPOINT 1 PTS

D is $ReCl_5$

CHECKPOINT 1 PTS

F is ReO_2

The gas **H** produced during the decomposition of $ReCl_5$ is Cl_2, because rhenium chlorides are not gaseous under normal heating conditions. Chlorine gas can be used for water purification, so statement 7 is correct.

CHECKPOINT 1 PTS

H is Cl_2

In summary, statements 2, 3, and 7 are correct, and statements 1, 4, 5, and 6 are incorrect. a is 62, b is 256, a/b = 0.242, B is correct.

CHECKPOINT 1 PTS

Statements 2, 3, and 7 are correct